

FIM590-003

Iowa House Price Prediction

Linear, Ridge, and Lasso Regression on a Log-Price Target

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Abstract

This project develops a reproducible linear-modeling workflow for Iowa house prices. The data contain 2,908 usable observations. Following the project specification, we construct 47 predictors, allocate 1,800 observations to training, 600 to validation, and 508 to a held-out test set, and compare ordinary least squares (OLS), three Ridge models, and three Lasso models. All models are estimated using $\log(1+\text{SalePrice})$ as the response. The validation set selects OLS with a validation mean squared error (MSE) of 0.014917, although Ridge with $\alpha = 0.60$ is nearly tied at 0.014929. After refitting OLS on the combined 2,400-observation development sample, the test-set results are MSE 0.014423, root mean squared error (RMSE) 0.120097, mean absolute error (MAE) 0.086159, and $R^2 = 0.908726$ on the log-price scale. The corresponding dollar-scale RMSE and MAE are approximately \$21,545 and \$14,997. Because the prescribed feature set contains exact linear dependencies, OLS predictions are valid but individual OLS coefficients are not uniquely identified. We therefore interpret coefficient patterns using the nearly tied Ridge model and treat all coefficients as predictive associations rather than causal effects.

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1 Project Objective

The objective is to predict Iowa residential sale prices with the 47-feature specification from the course material. The required workflow has four parts:

1. clean the original spreadsheet and construct the prescribed regressors;
2. create fixed training, validation, and test samples;
3. estimate OLS, Ridge, and Lasso models and select a model using validation MSE;
4. refit the selected specification on the development data and evaluate it once on the held-out test set.

The central modeling decision is to predict log price rather than dollar price. House prices are positive and right-skewed, so the transformation $\log(1 + y)$ reduces the influence of the most expensive houses and makes squared error closer to a proportional-error criterion. It also keeps the transformation defined at zero.

2 Data Preparation

2.1 Source data and cleaning

The source workbook contains 2,908 observations and 81 substantive columns after removing empty spreadsheet columns. Cleaning and feature construction follow these rules:

- strip whitespace from column names and remove unnamed empty columns;
- construct `BsmtFinSF` as `BsmtFinSF1 + BsmtFinSF2`;
- encode basement quality as `Ex = 5`, `Gd = 4`, `TA = 3`, `Fa = 2`, `Po = 1`, `NA = 0`;
- one-hot encode all 25 observed neighborhood categories;
- retain the 21 numerical variables, one basement-quality ordinal variable, and 25 neighborhood indicators listed in [Appendix A](#); and
- verify that the selected predictors and target contain no missing values.

The resulting design matrix has dimension 2908×47 . The intercept is estimated separately and is not counted among the 47 predictors.

2.2 Data split

A NumPy random generator with seed 20260928 produces one shuffled permutation of the observations. The first 1,800 observations form the training set, the next 600 form the validation set, and the remaining 508 form the test set. The test set is not used in model or penalty selection.

Table 1: Sample allocation and mean sale price.

Sample	Observations	Mean SalePrice (USD)
Training	1,800	180,018.88
Validation	600	178,133.52
Test	508	183,695.88

The similar mean prices in Table 1 indicate that the shuffled split did not produce an obvious price-level imbalance. Model selection nevertheless uses out-of-sample validation metrics rather than this descriptive comparison.

3 Modeling Methodology

3.1 Preprocessing and target transformation

Within each fit, the 21 numerical predictors and ordinal basement-quality variable are standardized using means and standard deviations learned only from that fit’s training sample. Neighborhood indicators remain 0/1. For a numerical feature x_j , the standardized value is

$$x_{ij}^* = \frac{x_{ij} - \bar{x}_j}{s_j}. \quad (1)$$

Let y_i be dollar sale price and let $\ell_i = \log(1 + y_i)$. The fitting pipeline also standardizes ℓ_i ,

$$z_i = \frac{\ell_i - \bar{\ell}}{s_\ell}, \quad (2)$$

and reverses both target transformations after prediction. Reported primary metrics are calculated on ℓ_i , not on the standardized target z_i . This distinction makes the reported MSE, RMSE, and MAE interpretable across fits.

3.2 Ordinary least squares

For design matrix X , response vector z , and intercept b_0 , OLS solves

$$\min_{b_0, \beta} \sum_{i=1}^n \left(z_i - b_0 - \mathbf{x}_i^\top \beta \right)^2. \quad (3)$$

When the columns are linearly independent, the normal-equation solution after centering is

$$\hat{\beta} = (X^\top X)^{-1} X^\top z. \quad (4)$$

The implemented numerical solver uses a least-squares decomposition and can also produce fitted values for a rank-deficient design. As discussed in Section 4.1, the fitted values can be well defined even when the individual coefficient vector is not unique.

3.3 Ridge regression

Ridge regression adds an L_2 penalty to shrink coefficients:

$$\min_{b_0, \beta} \|z - b_0 \mathbf{1} - X\beta\|_2^2 + \alpha \|\beta\|_2^2. \quad (5)$$

The intercept is not penalized. We estimate the three required values $\alpha \in \{0.10, 0.30, 0.60\}$. The penalty makes the coefficient solution unique and stable under the exact linear dependencies present in the feature set.

3.4 Lasso regression

Lasso uses an L_1 penalty and can set coefficients exactly to zero:

$$\min_{b_0, \beta} \frac{1}{2n} \|z - b_0 \mathbf{1} - X\beta\|_2^2 + \alpha \|\beta\|_1. \quad (6)$$

We estimate $\alpha \in \{0.02, 0.06, 0.10\}$ with a maximum of 50,000 coordinate-descent iterations. The loss scaling in Equation 6 follows the software implementation, so numerical α values should not be compared directly between Ridge and Lasso as if they represented identical penalty strength.

3.5 Evaluation metrics

For actual log prices ℓ_i , predicted log prices $\hat{\ell}_i$, and sample size m , the model-comparison statistics are

$$\text{MSE} = \frac{1}{m} \sum_{i=1}^m (\ell_i - \hat{\ell}_i)^2, \quad (7)$$

$$\text{RMSE} = \sqrt{\text{MSE}}, \quad (8)$$

$$\text{MAE} = \frac{1}{m} \sum_{i=1}^m |\ell_i - \hat{\ell}_i|, \quad (9)$$

$$R^2 = 1 - \frac{\sum_i (\ell_i - \hat{\ell}_i)^2}{\sum_i (\ell_i - \bar{\ell})^2}. \quad (10)$$

Validation MSE is the sole model-selection criterion. Dollar RMSE and MAE are reported only after selection as intuitive supplementary measures.

4 Validation Results and Model Selection

Table 2: Training and validation metrics on the log-price scale.

Model	Train MSE	Train RMSE	Train R^2	Val. MSE	Val. RMSE	Val. R^2
OLS	0.013908	0.117930	0.912059	0.014917	0.122137	0.914905
Ridge, $\alpha = 0.60$	0.013904	0.117914	0.912083	0.014929	0.122186	0.914837
Ridge, $\alpha = 0.30$	0.013902	0.117908	0.912092	0.014933	0.122201	0.914816
Ridge, $\alpha = 0.10$	0.013902	0.117906	0.912094	0.014936	0.122213	0.914798
Lasso, $\alpha = 0.02$	0.016669	0.129110	0.894595	0.017505	0.132305	0.900147
Lasso, $\alpha = 0.06$	0.019338	0.139063	0.877718	0.021760	0.147513	0.875871
Lasso, $\alpha = 0.10$	0.022969	0.151554	0.854762	0.026563	0.162982	0.848473

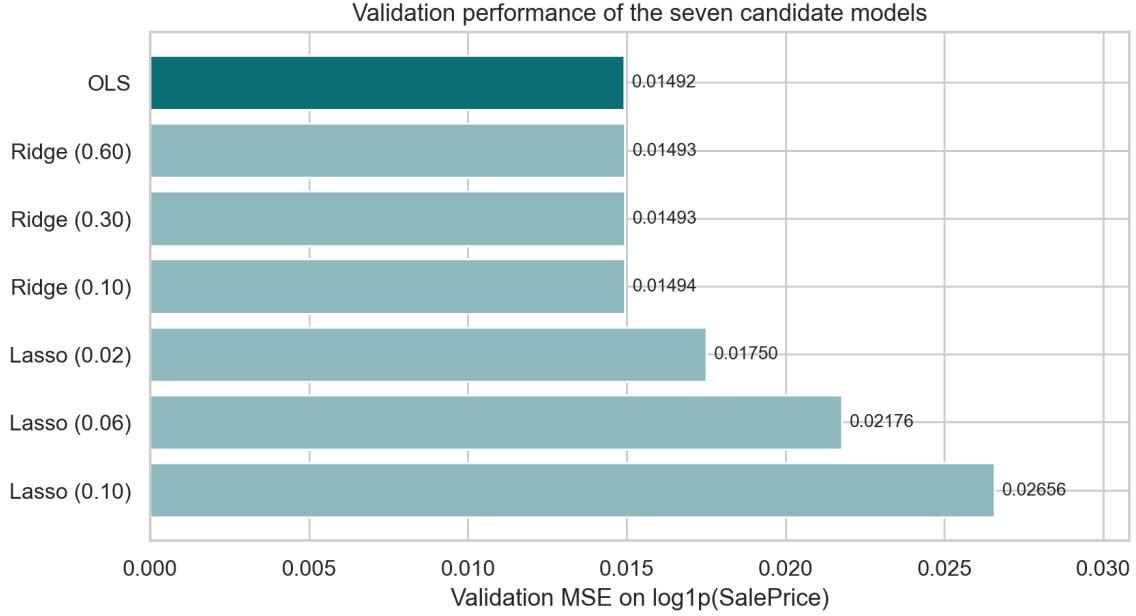


Figure 1: Validation MSE for all seven candidate models. Lower is better.

OLS has the smallest validation MSE and is therefore selected under the stated rule. Ridge with $\alpha = 0.60$ is only 0.000012 MSE units, or approximately 0.08%, above OLS. This is a practically negligible difference for this split. The Lasso models become progressively less accurate as α increases, which indicates that the stronger sparsity penalties discard useful predictive signal in this 47-feature specification.

The selected model has no regularization, so its method is OLS and its effective penalty is $\alpha = 0$. It is refitted on the combined training and validation sample of 2,400 observations before the single test evaluation.

4.1 Coefficient identification issue

The prescribed 47-feature matrix contains two exact dependencies:

$$\text{TotalBsmtSF} = \text{BsmtFinSF} + \text{BsmtUnfSF}, \quad (11)$$

$$\sum_{k=1}^{25} \text{Neighborhood}_{ik} = 1. \quad (12)$$

The second equality duplicates the intercept. Consequently, many different OLS coefficient vectors produce nearly the same fitted values. The numerical solution contains very large positive and negative coefficients that cancel, as shown in Appendix B. These values are a rank-deficiency artifact rather than evidence that a one-unit change in a feature changes log price by billions.

This issue does not invalidate the reported OLS prediction metrics. It does mean that the individual OLS slopes must not be interpreted. For a stable sensitivity view, Figure 4 uses Ridge with $\alpha = 0.60$, the closest validation competitor. A production model would normally remove one neighborhood indicator and one redundant basement-area variable, or use regularization.

5 Held-Out Test Performance

Table 3: Final performance of OLS refitted on 2,400 development observations.

Scale	Metric	Test value
$\log(1 + \text{SalePrice})$	MSE	0.014423
$\log(1 + \text{SalePrice})$	RMSE	0.120097
$\log(1 + \text{SalePrice})$	MAE	0.086159
$\log(1 + \text{SalePrice})$	R^2	0.908726
Dollars	RMSE	\$21,545.45
Dollars	MAE	\$14,996.77

The model explains approximately 90.9% of the variation in held-out log prices. An RMSE of 0.1201 log units corresponds roughly to a multiplicative error scale of $\exp(0.1201) - 1 \approx 12.8\%$, although RMSE is not itself a median percentage error. Dollar errors are larger for expensive houses, so the dollar RMSE exceeds the dollar MAE by about \$6,549.

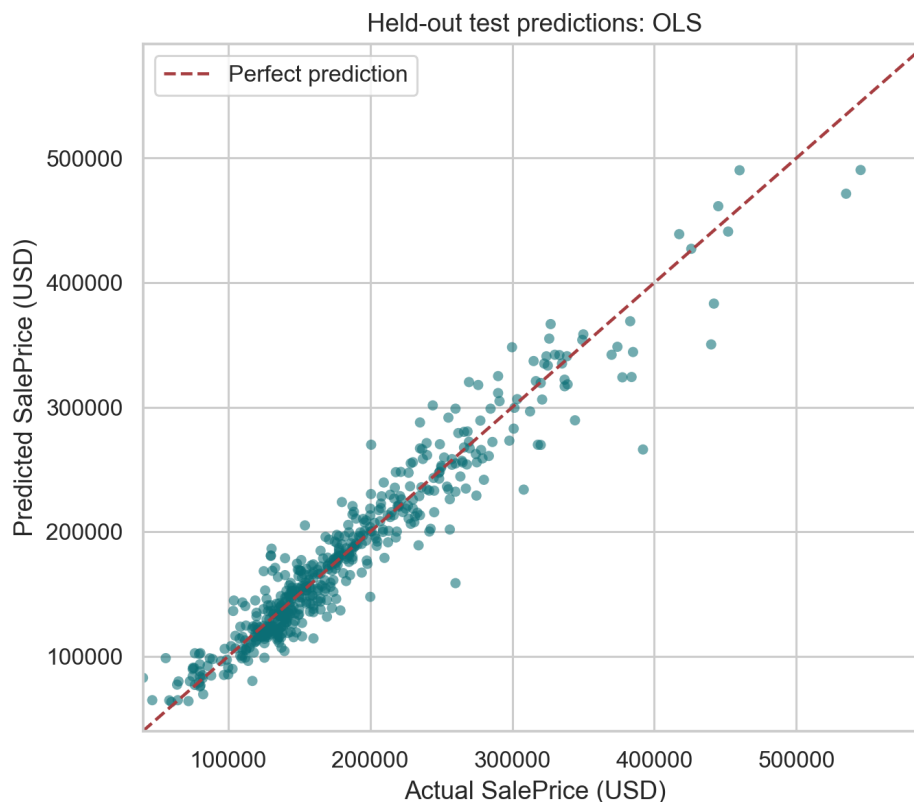


Figure 2: Actual and predicted sale prices for the 508 held-out test observations.

Figure 2 shows that most observations lie near the 45-degree reference line. The dispersion widens in dollar terms for expensive properties, which is consistent with modeling proportional rather than absolute dollar error.

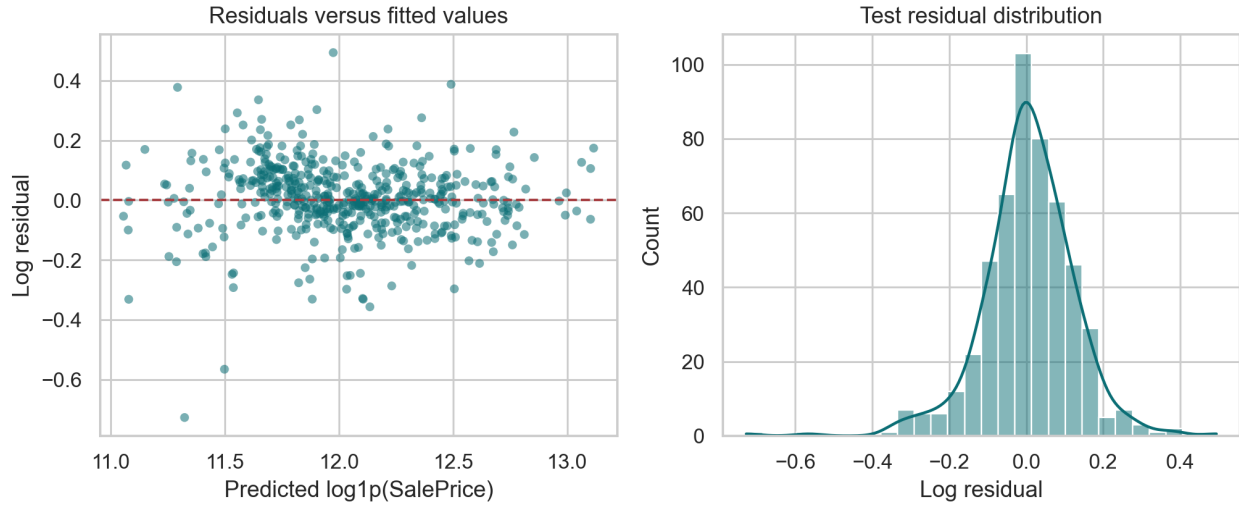


Figure 3: Test-set residual diagnostics on the log-price scale. Residual equals actual minus predicted log price.

The residual distribution is centered near zero, but several observations have large positive or negative errors. The residual-versus-fitted plot also suggests some nonconstant dispersion and local structure. These patterns are plausible for a linear model with a limited feature set and motivate the extensions discussed in Section 7.

6 Stable Coefficient Patterns

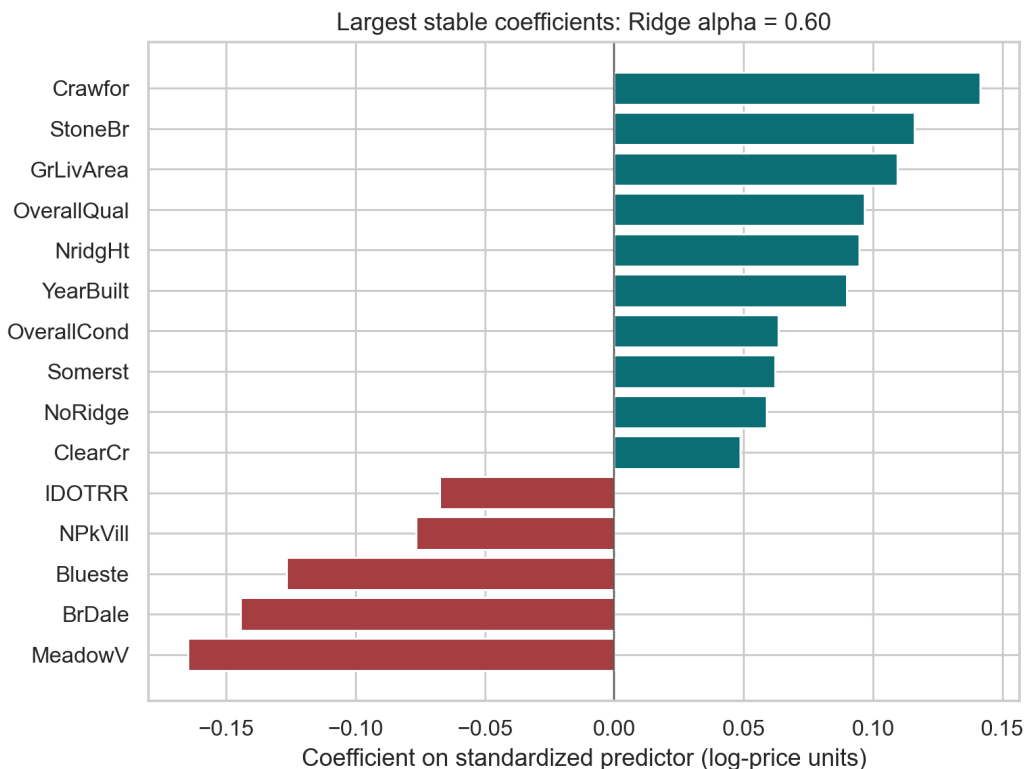


Figure 4: Largest standardized coefficients from Ridge with $\alpha = 0.60$, refitted on the 2,400-observation development sample.

The Ridge sensitivity model assigns large positive predictive associations to above-ground living area, overall quality, newer construction, and several neighborhood indicators, including Crawford, Stone Brook, and Northridge Heights. Meadow Village, Briardale, Bluestem, Northpark Villa, and Iowa DOT and Rail Road have negative associations conditional on the other included variables. The bars for standardized numerical predictors represent a one-standard-deviation change; neighborhood indicators remain 0/1. The magnitudes are therefore useful for pattern comparison but not identical unit changes across all feature types.

These are conditional predictive associations. They do not estimate causal effects of neighborhood, quality, renovation, or size because property characteristics and location are jointly determined and the design is observational.

7 Limitations and Possible Extensions

- **Single split.** Model ranking is based on one seeded validation split. The 0.08% validation-MSE gap between OLS and Ridge $\alpha = 0.60$ could change under another split. Repeated cross-validation would quantify this uncertainty.
- **Rank deficiency.** Exact feature dependencies make OLS coefficients non-unique. Reference-category encoding or removing a redundant basement variable would produce an identifiable OLS parameterization.

- **Linear functional form.** The model does not include interactions, splines, or nonlinear thresholds. Quality and size effects may vary across neighborhoods and price ranges.
- **Limited features.** The project deliberately uses 47 variables. Other available attributes, including additional exterior, condition, and location variables, may improve prediction.
- **Back-transformation.** Exponentiating a predicted conditional mean in log space does not generally equal the conditional mean price in dollars. A smearing correction could be considered when unbiased mean-dollar prediction is the goal.
- **External validity.** Results describe the supplied Iowa data and may not transfer to a different market or time period without retraining.

8 Conclusion

Using the required 47 features and fixed 1,800/600/508 split, OLS has the smallest validation MSE among the seven required candidates. After refitting on 2,400 development observations, it achieves test $R^2 = 0.9087$, log RMSE 0.1201, and dollar MAE of approximately \$15,000. Ridge with $\alpha = 0.60$ performs nearly identically on validation data and provides substantially more stable coefficients. The results support the usefulness of the log-price linear framework for prediction, while the rank-deficiency and residual diagnostics caution against interpreting the selected OLS coefficient vector as a set of unique economic effects.

A Complete List of 47 Features

Table 4: The 21 numerical features and one ordinal feature.

No.	Feature	Type	No.	Feature	Type
1	LotArea	Numerical	12	FullBath	Numerical
2	OverallQual	Numerical	13	HalfBath	Numerical
3	OverallCond	Numerical	14	BedroomAbvGr	Numerical
4	YearBuilt	Numerical	15	TotRmsAbvGrd	Numerical
5	YearRemodAdd	Numerical	16	Fireplaces	Numerical
6	BsmtFinSF	Numerical	17	GarageCars	Numerical
7	BsmtUnfSF	Numerical	18	GarageArea	Numerical
8	TotalBsmtSF	Numerical	19	WoodDeckSF	Numerical
9	1stFlrSF	Numerical	20	OpenPorchSF	Numerical
10	2ndFlrSF	Numerical	21	EnclosedPorch	Numerical
11	GrLivArea	Numerical	22	BsmtQual_ordinal	Ordinal

Table 5: The 25 neighborhood-indicator features.

No.	Feature	Type	No.	Feature	Type
23	NBHD_Blmngtn	Indicator	36	NBHD_NPkVill	Indicator
24	NBHD_Blueste	Indicator	37	NBHD_NWAmes	Indicator
25	NBHD_BrDale	Indicator	38	NBHD_NoRidge	Indicator
26	NBHD_BrkSide	Indicator	39	NBHD_NridgHt	Indicator
27	NBHD_ClearCr	Indicator	40	NBHD_OldTown	Indicator
28	NBHD_CollgCr	Indicator	41	NBHD_SWISU	Indicator
29	NBHD_Crawfor	Indicator	42	NBHD_Sawyer	Indicator
30	NBHD_Edwards	Indicator	43	NBHD_SawyerW	Indicator
31	NBHD_Gilbert	Indicator	44	NBHD_Somerst	Indicator
32	NBHD_IDOTRR	Indicator	45	NBHD_StoneBr	Indicator
33	NBHD_MeadowV	Indicator	46	NBHD_Timber	Indicator
34	NBHD_Mitchel	Indicator	47	NBHD_Veenker	Indicator
35	NBHD_NAmes	Indicator			

B Selected-Model Parameter Vector

Table 6 reports the solver’s parameter vector after refitting OLS on the 2,400 development observations. Parameters are expressed in original input units and log-price target units. The intercept is separate from the 47 slopes. Because the design is rank deficient, this is one numerical solution, not a unique coefficient vector. Predictions should be generated with the saved pipeline; the very large canceling coefficients must not be interpreted individually.

Table 6: One numerical parameter vector for the final OLS fit.

Parameter	Coefficient
Intercept	12093192747.369524
LotArea	2.3963165902358263e-06
OverallQual	0.070855827469600965
OverallCond	0.055650925487499212
YearBuilt	0.0028592520479720348
YearRemodAdd	0.00086390024081926787
BsmtFinSF	-767725470.55720007
BsmtUnfSF	-767725470.55728829
TotalBsmtSF	767725470.55737162
1stFlrSF	4.3564428622277209e-05
2ndFlrSF	2.3704389393927261e-05
GrLivArea	0.00022881150033979251
FullBath	0.0079451668564325569
HalfBath	0.0074616825071070758
BedroomAbvGr	-0.0077107208645243435
TotRmsAbvGrd	0.0036594772995293822
Fireplaces	0.031145754114527472
GarageCars	0.028188163713551193
GarageArea	0.00010946818198380031
WoodDeckSF	5.7800641971754614e-05
OpenPorchSF	9.7443749138376816e-05
EnclosedPorch	0.00015366988833127759
BsmtQual_ordinal	0.020465101325888978
NBHD_Blmngtn	-12093192744.249897
NBHD_Blueste	-12093192744.351295
NBHD_BrDale	-12093192744.378828
NBHD_BrkSide	-12093192744.199856
NBHD_ClearCr	-12093192744.173182
NBHD_CollgCr	-12093192744.207769
NBHD_Crawfor	-12093192744.087782
NBHD_Edwards	-12093192744.240747
NBHD_Gilbert	-12093192744.205662
NBHD_IDOTRR	-12093192744.301771
NBHD_MeadowV	-12093192744.399384
NBHD_Mitchel	-12093192744.239519
NBHD_NAmes	-12093192744.21439
NBHD_NPkVill	-12093192744.305983
NBHD_NWAmes	-12093192744.265966
NBHD_NoRidge	-12093192744.175354
NBHD_NridgHt	-12093192744.139254
NBHD_OldTown	-12093192744.277855
NBHD_SWISU	-12093192744.225033
NBHD_Sawyer	-12093192744.210047

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Parameter	Coefficient
NBHD_SawyerW	-12093192744.243877
NBHD_Somerst	-12093192744.172373
NBHD_StoneBr	-12093192744.124281
NBHD_Timber	-12093192744.198605
NBHD_Veenker	-12093192744.250326

C Reproducibility Algorithm

```

read and clean the spreadsheet
construct BsmtFinSF, ordinal BsmtQual, and 25 neighborhood indicators
assert 47 predictors and no missing selected values

permutation = random_generator(seed=20260928).permutation(2908)
train = permutation[:1800]
validation = permutation[1800:2400]
test = permutation[2400:]

for candidate in [OLS, Ridge(0.10, 0.30, 0.60), Lasso(0.02, 0.06, 0.10)]:
    fit preprocessing and target transformations on training data only
    fit candidate to standardized log1p(SalePrice)
    calculate training and validation metrics on the log-price scale

best_model = candidate with minimum validation MSE
refit best_model on training + validation data
evaluate once on the held-out test data

```

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